

## Environment and Preferences

A two-period OLG economy:  $Y_t$  young and  $O_t$  old share a fixed housing stock  $\bar{H}$ .

- Entrants draw income and liquid wealth,  $(y_i, b_i)$ , from a fixed distribution  $F$ .
- Young choose consumption, housing, fertility, saving, and tenure.
- Old choose consumption, housing, and an estate.

Flow utilities are

$$u_t^Y(c, h, n) = \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta_t \log n,$$

$$u^2(c^2, h^2, e) = \log c^2 + \gamma \log h^2 + \omega_B \log e.$$

Each child needs  $\chi$  goods and  $\kappa$  housing;  $e$  gives warm-glow utility.

Old-age utility is discounted by  $\beta$ ; owning adds the taste  $\xi_i$ .

## Housing and Finance

At house price  $P_t$ , a mortgage finances share  $\phi_t$ . Liquid wealth  $b_i$  funds the down payment:

$$\underbrace{h \leq h_R^{\max} < h_O^{\max}}_{\text{rent}} \qquad \underbrace{h \leq h_O^{\max}, \quad (1 - \phi_t)P_t h \leq b_i}_{\text{own}}.$$

A world bond has price  $q$  and gross return  $R_f = q^{-1}$ . With next-period rent  $r_t$  and property-tax rate  $\tau^P$ , rental entry implies

$$qr_t = (1 + q\tau^P)P_t - qP_{t+1} > 0.$$

- Income and rebates arrive after closing; unsecured borrowing is unavailable.
- Old owners may downsize, but cannot buy more or let retained housing.

## Competitive Equilibrium

Given initial cohorts and old households' assets, households choose tenure and allocations taking the full price path as given.

Housing clears; property-tax revenue is rebated equally as  $T_t$ :

$$Y_t \bar{h}_t^Y + O_t \bar{h}_t^O = \bar{H}, \quad (Y_t + O_t) T_t = q \tau^P P_t \bar{H}.$$

Here  $\bar{h}_t^Y, \bar{h}_t^O$  are average housing by age, across both tenures.

Mean completed fertility  $\bar{n}_t$  determines the next young cohort:

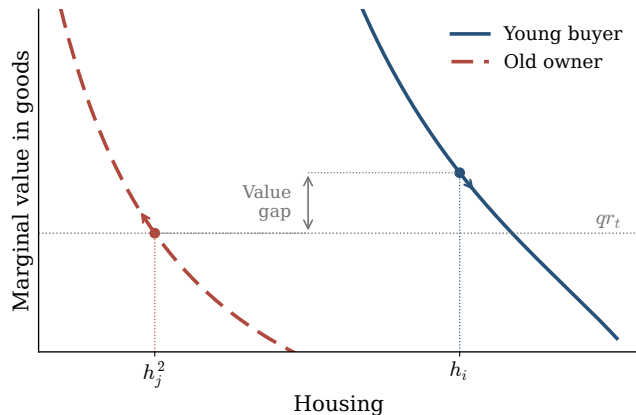
$$Y_{t+1} = \nu \bar{n}_t Y_t, \quad O_{t+1} = Y_t,$$

where  $\nu$  converts children into entering households.

In a positive steady state,  $Y^* = O^*$  and  $\bar{n}^* = 1/\nu$ .

# Housing Misallocation

Marginal housing values are measured in units of current consumption.



$$MV_i^Y = \frac{\alpha(c_i - \chi n_i)}{h_i - \kappa n_i}$$

$$MV_j^O = \frac{\gamma c_j^2}{h_j^2}$$

$$MV_i^Y > q_r_t = MV_j^O.$$

Young down payments bind; saving is positive; owner size, retention, and estate limits are slack.

## A Compensated Reallocation

Move a little housing from old owners to young owners with a higher marginal value.

- Compensate the old with goods from the young.
- Keep fertility, tenure, estates, and future real allocations fixed.
- Respect physical limits and total resources; relax private financing limits.

**Proposition.** A sufficiently small reallocation makes the young better off and leaves everyone else equally well off.

The value gap holds on positive masses of feasible pairs. There is no real moving cost, and existing financial claims are honored.

## Housing and Fertility

Consider a young buyer with a binding housing limit, positive saving, and interior old-age choices. Hold prices and tenure fixed.

When the housing limit is relaxed, fertility rises if and only if

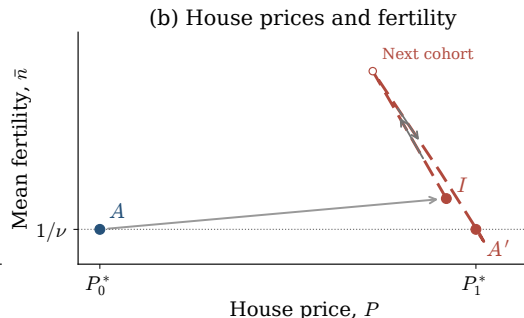
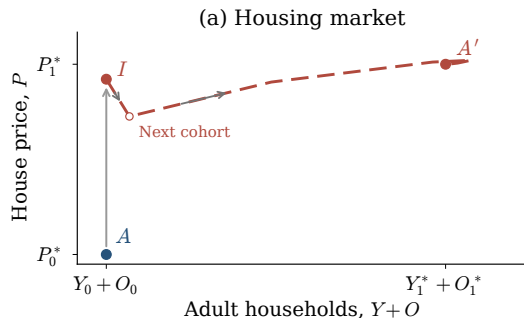
$$\underbrace{\frac{\alpha\kappa}{(h - \kappa n)^2}}_{\text{space benefit}} > \underbrace{\frac{\chi q r_t}{[1 + \beta(1 + \gamma + \omega_B)](c - \chi n)^2}}_{\text{resource cost}}.$$

More space helps accommodate children; paying for it uses other resources.

Saving and old-age choices adjust; the same constraints bind. No accompanying transfers.

# Equilibrium Transition

After a small permanent credit relaxation,  $\phi_1 > \phi_0$ :



$A$ : initial steady state;  $I$ : impact;  $A'$ : new steady state.

A local analytical example with both tenures. Fertility returns to replacement, with a larger population. Adjustment can overshoot.